DCCL: Dual-View Cross-Modal Consistency Learning for Hyperspectral and LiDAR Data Classification Under Noisy Labels

Xia Yue, Anfeng Liu, Ning Chen, Shaobo Xia, Jun Yue, *Member, IEEE*, and Leyuan Fang, *Senior Member, IEEE*

***Abstract*—Joint hyperspectral and light detection and ranging (LiDAR) data classification shows great promise for object recognition and land-cover mapping. However, most existing approaches assume clean labels—an assumption that may not hold in practical remote sensing applications—leaving noisy-label learning for real-world hyperspectral image (HSI)–LiDAR classification largely unexplored. To address this gap, we formulate the problem of HSI–LiDAR data classification under noisy labels and establish a baseline. We further develop a dual-view cross-modal consistency learning (DCCL) framework to enhance robustness against both symmetric and asymmetric label noise. In particular, we introduce a dual-view augmentation strategy that applies strong and weak perturbations separately to HSI and LiDAR data, providing label-independent supervisory signals and encouraging perturbation-invariant and modality-coherent representations. Furthermore, we propose a cross-modal consistency learning strategy that enforces prediction consistency across views and representation consistency across modalities and views, enabling robust dual-modal feature learning under noisy labels. Experiments on the Houston 2013, Trento, and MUUFL datasets under both symmetric and asymmetric noisy-label settings demonstrate that the proposed method consistently outperforms existing approaches. The results highlight the vulnerability of current HSI–LiDAR classification methods to label noise and confirm the effectiveness of the proposed DCCL framework. The source code will be made publicly available at https://github.com/halfcoder1/DCCL**

***Index Terms*—Consistency learning, hyperspectral image (HSI), light detection and ranging (LiDAR) data, multimodal learning, noisy labels.**

Received 7 September 2026; revised 29 September 2026; accepted 9 October 2026. Date of publication 13 October 2026; date of current version 22 October 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62535017, Grant 62575039, and Grant 62575311; and in part by Hunan Provincial Natural Science Foundation of China under Grant 2025JJ40059. *(Corresponding authors: Anfeng Liu; Shaobo Xia.)*

Xia Yue is with the School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (e-mail: yuexia486@gmail.com).

Anfeng Liu and Shaobo Xia are with the School of Electronic Information, Central South University, Changsha 410083, China (e-mail: afengliu@mail.csu.edu.cn; shaoboxia2020@gmail.com).

Ning Chen is with the Institute of Remote Sensing and Geographic Information System, School of Earth and Space Sciences, Peking University, Beijing 100871, China (e-mail: chenning0115@pku.edu.cn).

Jun Yue is with the School of Automation, Central South University, Changsha 410083, China (e-mail: junyue@csu.edu.cn).

Leyuan Fang is with the School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China, and also with the Pengcheng Laboratory, Shenzhen 518000, China (e-mail: fangleyuan@gmail.com).

Digital Object Identifier 10.1109/TGRS.2026.3743120

I. Introduction

HYPERSPECTRAL image (HSI) provides data cubes with high spectral resolution, playing a pivotal role in scene classification, component retrieval, and land-cover mapping [1]. Light detection and ranging (LiDAR) captures accurate 3-D structural information—particularly elevation—which serves as a powerful complement to HSIs. Beyond providing 3-D spatial priors for inversion and reconstruction, LiDAR effectively resolves spectral ambiguities by introducing height as an additional discriminative dimension [2], thereby facilitating the separation of spectrally similar but height-distinct classes (e.g., grass versus trees, or roads versus rooftops).

Fusing these two heterogeneous modalities is far from trivial, necessitating not only specialized network architectures but also highly reliable supervisory signals to ensure effective multimodal interpretation. Recent studies have largely focused on developing dual-modal fusion architectures. For example, Zhao et al. [3] employs a dual-branch design in which a hierarchical CNN extracts local spatial–spectral cues while a Transformer models long-range semantics and fuses HSI–LiDAR features via cross-token attention. Liao et al. [4] integrates CNNs for local detail extraction with Mamba for sequence-level dependency modeling, jointly embedding HSI spectral–spatial features, LiDAR elevation, and LiDAR-derived edge cues to overcome the difficulty of capturing both fine details and long-range geometric relationships in complex scenes. While most existing HSI–LiDAR fusion studies concentrate on network design, far less attention has been paid to the reliability of the supervisory signals. When labels are noisy, the complementary strengths of the two modalities may fail to emerge and may even mislead the model, ultimately producing distorted feature representations, weakened fusion benefits, and notable performance degradation. In particular, noisy supervision introduces unreliable gradients that may distort the decision boundary and propagate errors across modalities through feature fusion. These difficulties call for learning constraints that are less dependent on potentially corrupted annotations and can preserve stable cross-modal representations.

In real-world applications, acquiring large-scale and high-quality annotations for remote sensing imagery is challenging, particularly for HSI [5], [6]. The difficulties arise mainly from two factors.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

1) Hyperspectral annotation is costly and error-prone, as it often depends on labor-intensive in situ surveys that are unavoidably affected by measurement uncertainties.
2) Spatiotemporal variability and inconsistent label quality across different sources (e.g., experts, crowd-sourcing, and historic maps) preclude seamless correspondence between annotations and multimodal remote sensing data.

For instance, [7] reported that even in relatively simple scenes containing only six easily identifiable categories, human annotation errors exceeded 17%, while in complex or ambiguous environments, error rates for some classes surpassed 30%. In addition, [8], [9] have revealed geometric offset errors typically ranging from 5.0 to 10.0 m in certain annotations. Compared with these cases, HSI involves more diverse land-cover categories, finer discrimination requirements, and greater dependence on expert knowledge. As a result, fine-grained annotation for HSI is especially prone to noisy labels, which in turn severely hinder reliable HSI–LiDAR fusion and interpretation.

Semantic label noise should be distinguished from observation degradation. Gaussian noise, deadlines, stripes, impulse noise, and their mixtures corrupt hyperspectral measurements and are addressed by image restoration or joint denoising and classification methods [10]. On the contrary, label noise concerns incorrect semantic annotations associated with the observed samples. Its distribution depends on the annotation process: ambiguous or mixed pixels may exhibit instance-dependent errors, confusion between similar categories may produce class-dependent errors, and annotation-map offsets or temporal changes may induce spatially correlated errors. Accordingly, no single synthetic distribution fully represents practical HSI–LiDAR label noise. The symmetric and cyclic asymmetric models adopted here provide controlled evaluations rather than complete descriptions of real-world annotation errors.

Consequently, noisy-label learning for HSI interpretation has attracted increasing research interest. Tu et al. [11] tackled random label noise arising from mixed pixels, complex land-cover boundaries, and human annotation errors by introducing a superpixel–pixel weighting distance-based density estimator to identify and remove mislabeled samples. Wang et al. [12] addressed the coexistence of noisy labels and severe class imbalance through bidirectional curriculum learning and stacking ensemble. Noisy-label learning has also been explored for LiDAR interpretation. Gao et al. [13] developed a dynamic class-wise label correction strategy based on the class-dependent separability of learning curves. However, these methods are designed for a single modality, either HSI or LiDAR, and do not account for the interaction between noisy supervision and multimodal fusion. Consequently, extending noisy-label learning to HSI–LiDAR classification remains largely unexplored.

The limited exploration of multimodal noisy-label learning arises from both practical and methodological difficulties. From a practical perspective, most existing HSI–LiDAR benchmarks and fusion methods are developed under the assumption of clean paired annotations, while multimodal datasets and evaluation protocols explicitly considering noisy supervision remain scarce. From a methodological perspective, conventional noisy-label methods are generally designed for a single feature space and cannot directly handle the interaction between label corruption and heterogeneous multimodal representations. In particular, three key challenges must be addressed. First, because coregistered HSI and LiDAR observations share the same observed label, an incorrect annotation can simultaneously bias both modality-specific branches toward the same erroneous semantic target. Second, multimodal fusion may transfer and amplify noise-induced errors across modalities, resulting in more severe error propagation than in single-modality learning. Third, cross-modal agreement must be imposed carefully, since overly strong consistency may reinforce incorrect predictions, whereas insufficient alignment may fail to exploit complementary spectral and structural information. These difficulties require a unified framework that balances noise robustness, cross-modal complementarity, and representation consistency.

To address these challenges, we propose a dual-view cross-modal consistency learning (DCCL) framework for robust HSI and LiDAR classification under noisy labels. DCCL constructs modality-specific weak and strong views and introduces complementary consistency constraints at both the prediction and representation levels. In particular, dual-view prediction consistency (DVPC) stabilizes model outputs across augmentation strengths, cross-modal representation contrast (CMRC) aligns semantically corresponding HSI and LiDAR representations, and view-invariant representation contrast (VIRC) preserves instance-level semantics under perturbations. Together, these designs reduce dependence on corrupted labels and suppress the propagation and reinforcement of noise-induced errors during multimodal fusion. The primary contributions of this article are summarized as follows.

1) To the best of our knowledge, we are the first to investigate joint HSI and LiDAR classification under noisy labels, extending noise-robust learning from conventional single-modality settings to multimodal remote sensing. We further conduct systematic evaluations under both symmetric and asymmetric label noise to assess the robustness of multimodal classification methods.
2) We develop a modality-specific dual-view augmentation strategy that constructs weakly and strongly perturbed views for both HSI and LiDAR data. The weak views preserve semantic content, whereas the strong views introduce label-preserving spectral and elevation perturbations, providing complementary views for label-independent consistency learning and promoting perturbation-invariant multimodal representations.
3) We propose a unified consistency learning framework that integrates DVPC, CMRC, and VIRC. These complementary objectives enforce prediction stability across views, semantic alignment between heterogeneous modalities, and instance-level invariance in both unimodal and fused representation spaces, thereby suppressing noise propagation during multimodal fusion and improving classification robustness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

The remainder of this article is organized as follows. Section II reviews related work on HSI–LiDAR data fusion and learning with noisy labels. Section III presents the proposed framework. Section IV presents the experimental results and analysis. Section V concludes this article.

II. Related Work

A. HSI and LiDAR Data Classification

HSI classification is a fundamental task in remote sensing that assigns a semantic land-cover label to each pixel based on its high-dimensional spectral signature [14]. Owing to its fine spectral resolution, HSI enables precise material discrimination and has been widely applied to urban mapping [15], agricultural monitoring [16], and environmental analysis [17]. Nevertheless, its high dimensionality, limited labeled samples, and complex spectral–spatial variability pose considerable challenges to robust classification [18], [19].

To alleviate these limitations, complementary sensing modalities have been introduced [20], [21]. In particular, LiDAR provides elevation and geometric information that can compensate for the structural ambiguity of hyperspectral imagery [22], [23]. This complementarity has motivated the joint exploitation of HSI and LiDAR data for more reliable land-cover classification [24], [25], [26], [27]. Most existing HSI–LiDAR classification methods rasterize or project LiDAR measurements onto the HSI image grid and perform fusion in a coregistered 2-D space [28]. By jointly exploiting spectral–spatial and structural information, these methods improve class separability, particularly for categories with similar spectral signatures but different elevation characteristics [29], [30], [31].

Recent studies have also investigated spectral–geometric complementarity under different acquisition conditions and data representations. For example, [32] integrated multitemporal airborne HSI and LiDAR data for city-scale urban tree species classification, exploiting both seasonal and multimodal information. Beyond raster-based fusion, Liu et al. [33] developed an adaptive multiscale fusion method for multispectral point-cloud classification, addressing sparse annotations, variations in land-cover scale, and long-tailed class distributions. Jiang et al. [34] proposed a pillar–voxel fusion network for 3-D object detection in airborne hyperspectral point clouds, mitigating geometric–spectral distortions associated with fusion errors and occlusions.

Overall, these studies exploit complementary spectral and geometric information through diverse fusion architectures and data representations. Robust learning with unreliable semantic annotations constitutes a related but distinct challenge, which is reviewed in Section II-B.

B. HSI and LiDAR Data Classification With Noisy Labels

A growing body of research has investigated noise-robust HSI classification, primarily in single-modality settings [35], [36]. Existing approaches exploit neighborhood consistency to select reliable samples [37], employ contrastive learning and low-confidence sample reutilization to improve spectral–spatial robustness [38], identify noisy regions through uncertainty modeling [39], or estimate source-specific noise patterns when multiple heterogeneous annotations are available [40]. Although these methods reduce the adverse influence of corrupted labels, they are designed for a single spectral–spatial feature space and do not consider error interactions between heterogeneous modalities.

Noise-robust learning has also been explored for LiDAR point-cloud interpretation. Representative methods perform point-wise confidence selection and cluster-wise label correction [41] or suppress unreliable supervision through graph-based prototype modeling and transductive label propagation [42]. However, these approaches are developed for single-modality LiDAR data and do not consider joint representation learning with HSI.

The distinction between DCCL and existing studies lies in both the learning setting and the regularization design. The conventional HSI–LiDAR classification methods reviewed above primarily emphasize fusion architectures and cross-modal feature integration, whereas noisy-label methods for HSI or LiDAR focus on robust learning within a single sensing modality. Multimodal noisy-label learning has also been investigated. For example, CromSS [43] uses cross-modal sample selection and spatial–temporal label smoothing for noisy-label pretraining and subsequent transfer to remote sensing segmentation tasks. Other multimodal robustness studies address different sources of uncertainty. For instance, the method in [44] filters irrelevant visual information and summarizes attributes for multimodal entity alignment, focusing on modality-content noise rather than corrupted land-cover annotations.

On the contrary, DCCL targets joint HSI–LiDAR pixel classification with corrupted semantic labels shared by the paired modalities. Its distinguishing feature is the combination of modality-specific weak and strong views with three complementary objectives: DVPC regularizes predictions across views, CMRC aligns HSI and LiDAR embeddings, and VIRC maintains instance-level consistency across views in both unimodal and fused feature spaces. Compared with the noisy-label pretraining and sample-selection strategy of CromSS, DCCL incorporates these objectives directly into the training of the HSI–LiDAR classifier. This design jointly regularizes predictions and representations to improve classification robustness under noisy supervision.

III. Method

As illustrated in Fig. 1, DCCL combines modality-specific dual-view augmentation with DVPC, CMRC, and VIRC to regularize predictions and representations under noisy supervision. Sections III-A–III-G present the problem formulation, augmentation strategy, learning objectives, and computational complexity.

A. Problem Formulation

Let Y and $\tilde{Y}$ denote the random variables corresponding to clean and noisy labels, respectively, and let y and $\tilde{y}$ be their realizations at the pixel level. Label corruption is modeled by a transition matrix $T \in \mathbb{R}^{K\times K}$, where K denotes the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

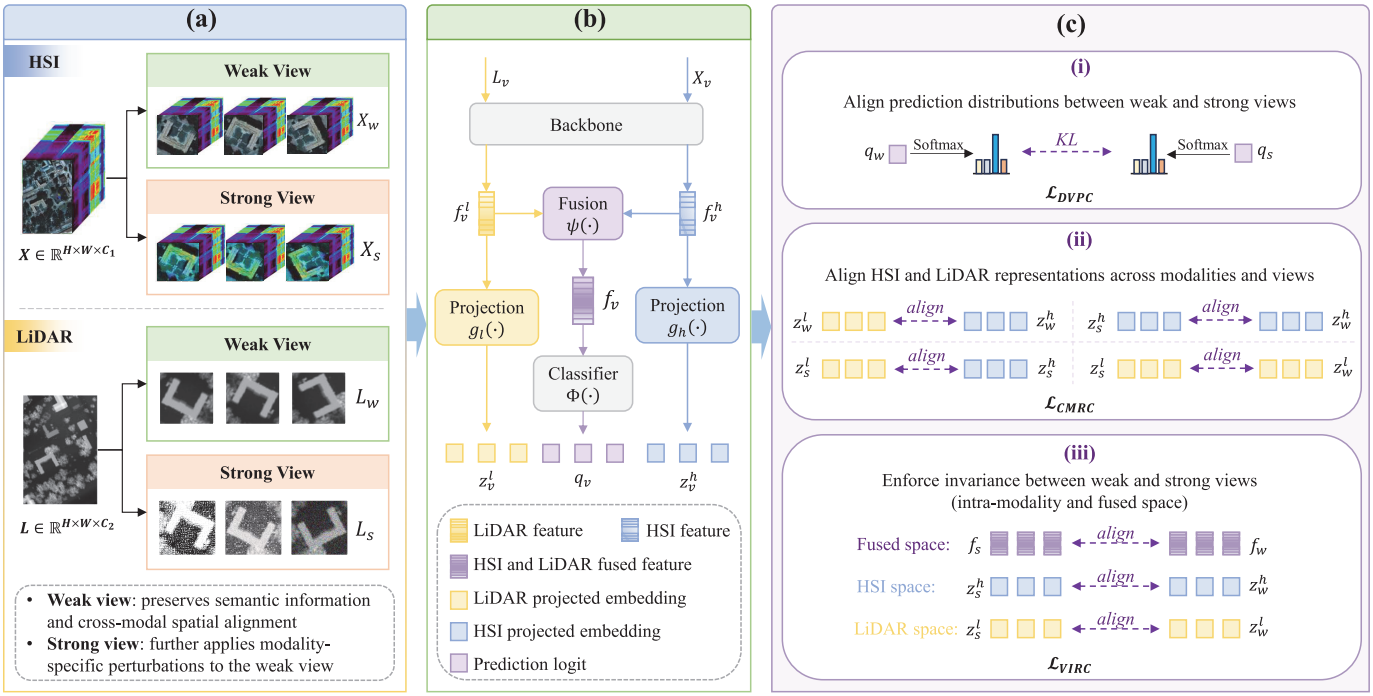


Fig. 1. Overview of the proposed DCCL framework. Given HSI and LiDAR inputs, modality-specific weak and strong views are first constructed through dual-view multimodal augmentation, where the weak view preserves semantic information and cross-modal spatial alignment, while the strong view introduces modality-specific label-preserving perturbations. The augmented views are then processed by shared modality-specific backbones to extract HSI and LiDAR representations, followed by feature fusion and classification. Meanwhile, modality-specific projection heads map unimodal features into embedding spaces for contrastive learning. The proposed consistency learning objectives include DVPC, which aligns prediction distributions between weak and strong views; CMRC, which aligns HSI and LiDAR embeddings across modalities and views; and VIRC, which enforces invariant representations within individual modalities and the fused feature space. (a) Dual-view multimodal augmentation. (b) Feature extraction. (c) Consistency and contrastive learning objectives. (i) DVPC. (ii) CMRC. (iii) VIRC.

number of semantic classes and $T_{y\tilde{y}} = p(\tilde{Y} = \tilde{y} \mid Y = y)$. We consider symmetric and asymmetric noise models, which impose different structures on T.

In particular, in the symmetric noise model, a clean label is uniformly flipped to any other class with probability η_{sym}

$$T_{y\tilde{y}} = \begin{cases} 1 - \eta_{\text{sym}}, & \tilde{y} = y \\ \dfrac{\eta_{\text{sym}}}{K-1}, & \tilde{y} \neq y. \end{cases} \tag{1}$$

In the asymmetric noise model, label corruption follows a class-dependent single-target transition pattern: each clean class y can be flipped only to a predefined target class $\pi(y) \neq y$. The corresponding transition probabilities are defined as follows:

$$T_{y\tilde{y}} = \begin{cases} 1 - \eta_{\text{asym}}, & \tilde{y} = y \\ \eta_{\text{asym}}, & \tilde{y} = \pi(y) \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

In our experiments, we use a cyclic mapping $\pi(y) = (y \bmod K) + 1$, with class indices ranging from 1 to K. Following the class ordering in Table I, the mapping is $1 \to 2, \ldots, K-1 \to K, K \to 1$. Together with (2), this mapping fully specifies the asymmetric noise transition matrix for each dataset.

These settings provide complementary controlled evaluations: symmetric noise distributes corruption uniformly across the other classes, whereas cyclic asymmetric noise concentrates corruption from each class on a single target. The latter evaluates robustness to structured, class-dependent label corruption. However, the cyclic mapping is based on class indices rather than spectral–structural similarity or empirical annotation errors and therefore does not necessarily represent physically plausible confusion between land-cover categories.

Let $\mathbf{p}_Y$ and $\mathbf{p}_{\tilde{Y}}$ denote the K-dimensional column vectors of clean and noisy class probabilities, respectively. Under the class-dependent corruption model, they satisfy

$$\mathbf{p}_{\tilde{Y}} = T^{\top} \mathbf{p}_Y. \tag{3}$$

The transition matrix T is used only to specify and generate synthetic label noise. It is not provided to the classifier or used to correct its predictions or training loss.

Given paired HSI and LiDAR observations $(\mathbf{x}^h, \mathbf{x}^l)$ and their observed noisy labels $\tilde{y}$, our goal is to learn a classifier $f_\theta(\cdot)$ that predicts the underlying clean classes. The classifier outputs a K-dimensional class probability vector after softmax normalization. The basic supervised risk based on the observed labels is written as follows:

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}^h, \mathbf{x}^l, \tilde{y})} \left[\mathcal{L}_{\text{CE}} \left(f_\theta \left(\mathbf{x}^h, \mathbf{x}^l \right), \tilde{y} \right) \right] \tag{4}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss. Directly minimizing this risk may lead to overfitting to corrupted labels. DCCL therefore incorporates mixup and label-independent consistency regularization to mitigate this effect. The complete training objective is detailed in Section III-F.

The noise models in (1)–(3) describe corruption in the semantic label space. We assume that HSI and LiDAR are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I

CLASS NAMES AND THE NUMBERS OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON 2013, TRENTO, AND MUUFL DATASETS. FOR ALL DATASETS, 50 SAMPLES PER CLASS ARE USED FOR TRAINING

Houston2013				Trento				MUUFL			
No.	**Name**	**Train**	**Test**	**No.**	**Name**	**Train**	**Test**	**No.**	**Name**	**Train**	**Test**
1	Healthy grass	50	1201	1	Apple trees	50	3984	1	Trees	50	23196
2	Stressed grass	50	1204	2	Buildings	50	2853	2	Mostly grass	50	4220
3	Synthetic grass	50	647	3	Ground	50	429	3	Mixed ground surface	50	6832
4	Trees	50	1194	4	Wood	50	9073	4	Dirt and sand	50	1776
5	Soil	50	1192	5	Vineyard	50	10451	5	Road	50	6637
6	Water	50	275	6	Roads	50	3124	6	Water	50	416
7	Residential	50	1218					7	Building shadow	50	2183
8	Commercial	50	1194					8	Buildings	50	6190
9	Road	50	1202					9	Sidewalk	50	1335
10	Highway	50	1177					10	Yellow curb	50	133
11	Railway	50	1185					11	Cloth panels	50	219
12	Parking Lot 1	50	1183								
13	Parking Lot 2	50	419								
14	Tennis Court	50	378								
15	Running Track	50	610								
Total		**750**	**14279**	**Total**		**300**	**29914**	**Total**		**550**	**53137**

coregistered to a common spatial grid, such that each paired sample $(\mathbf{x}^h, \mathbf{x}^l)$ shares a single observed noisy label $\tilde{y}$. We therefore do not simulate separate label-noise processes for the two modalities. In practice, displacement of an annotation map relative to the observations may assign incorrect semantic labels to pixels, particularly near class boundaries. By contrast, misregistration between HSI and LiDAR may cause $\mathbf{x}^h$ and $\mathbf{x}^l$ to represent different physical locations, making a label that is correct for one modality inconsistent with the other. Our experiments introduce class-label corruption without artificially shifting the annotation map or either modality. Accordingly, the evaluation focuses on robustness to semantic label corruption under the coregistration assumption; robustness to geometric misregistration remains to be investigated.

B. Dual-View Multimodal Augmentation

Given an HSI cube $X \in \mathbb{R}^{H\times W\times C_1}$ and a coregistered LiDAR elevation map $L \in \mathbb{R}^{H\times W\times C_2}$, we extract spatial patches P^h_{ij} and P^l_{ij} centered at spatial location (i, j), where (i, j) ranges over all valid positions that allow a patch of predefined size $P \times P$ to be sampled. Here, C_1 and C_2 denote the numbers of spectral bands and LiDAR channels, respectively, and H and W correspond to the spatial height and width. To facilitate noise-robust representation learning, we generate two augmented views for each multimodal patch following a dual-view augmentation strategy. Here, "dual-view" refers to two differently augmented versions—weak and strong—of the same spatially aligned HSI–LiDAR patch.

The *weak augmentation* applies geometry-preserving transformations, including random horizontal and vertical flips and rotations by 90°, 180°, or 270°. These transformations are applied jointly to the HSI patch, the LiDAR patch, and the corresponding label map to maintain spatial alignment across modalities. The *strong augmentation* introduces additional modality-specific perturbations on top of the same geometric transformation. For the HSI modality, a small random spectral shift is applied by rolling the spectral dimension within a predefined limited range. This operation simulates minor band misalignment or calibration drift while largely preserving the local spectral structure, thereby encouraging robustness to spectral variations. For the LiDAR modality, Gaussian noise is injected to perturb elevation values, mimicking measurement uncertainty and sensor-level variations. The weak–strong augmentation design follows the consistency-based learning paradigm widely adopted in contrastive and semisupervised learning (e.g., SimCLR [45] and FixMatch [46]-style frameworks), where the weak view provides stable semantic guidance and the strong view introduces perturbation-based invariance constraints.

We emphasize that these perturbations are designed for augmentation-driven invariance learning rather than label-noise simulation. As a result, two augmented multimodal views are obtained, denoted as (X_w, L_w) and (X_s, L_s), corresponding to the weak and strong views, respectively. Both views are processed by shared modality-specific encoders and a shared fusion module, providing representations for the consistency objectives described below.

C. Dual-View Prediction Consistency

Based on the dual-view augmented inputs introduced in Section III-B, we now formalize the multimodal prediction pipeline for weakly and strongly augmented views, which serves as the foundation for the proposed prediction consistency regularization.

For notational convenience, we denote the augmented inputs as (X_v, L_v) with $v \in \{w, s\}$. The HSI encoder $\phi_h(\cdot)$ and the LiDAR encoder $\phi_l(\cdot)$ are two modality-specific feature extractors defined as follows:

$$\phi_h : \mathbb{R}^{P\times P\times C_1} \to \mathbb{R}^{d_h}, \qquad \phi_l : \mathbb{R}^{P\times P\times C_2} \to \mathbb{R}^{d_l} \tag{5}$$

where d_h and d_l denote the feature dimensions of the HSI and LiDAR representations, respectively. Given an augmented view v, the corresponding unimodal features are computed as follows:

$$f^h_v = \phi_h(X_v), \qquad f^l_v = \phi_l(L_v), \quad v \in \{w, s\} \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

where f_v^h and f_v^l denote the HSI and LiDAR feature representations extracted from view v, respectively. A fusion network $\psi(\cdot)$ integrates the unimodal features to produce a multimodal representation

$$f_v = \psi\left(\left[f_v^h, f_v^l\right]\right) \in \mathbb{R}^d \tag{7}$$

where $[\cdot,\cdot]$ denotes feature concatenation and d is the dimension of the fused embedding. In particular, we follow the fusion design of S2ENet [47] to construct the multimodal representation. A linear classifier is subsequently applied to f_v to predict class probabilities.

For the representation-level contrastive objectives, two independent modality-specific projection heads, $g_h(\cdot)$ and $g_l(\cdot)$, map the HSI and LiDAR features into a common-dimensional embedding space

$$z_v^h = g_h\left(f_v^h\right), \qquad z_v^l = g_l\left(f_v^l\right), \qquad v \in \{w, s\}. \tag{8}$$

Both projection heads are implemented as two-layer multilayer perceptrons (MLPs) with an output dimension of 32. They share the same architecture but have separate learnable parameters, while each head is shared across the weak and strong views of its corresponding modality. The network produces the corresponding prediction logits

$$q_w = \Phi_{\text{cls}}(f_w), \qquad q_s = \Phi_{\text{cls}}(f_s)$$

where $\Phi_{\text{cls}}(\cdot)$ denotes the shared linear classification head. We enforce prediction consistency across dual views via a KL-based regularization

$$\mathcal{L}_{\text{dvpc}} = \mathbb{E}\left[\text{KL}\left(\text{softmax}\left(\frac{q_w}{\tau}\right) \,\middle\|\, \text{softmax}\left(\frac{q_s}{\tau}\right)\right)\right] \tag{9}$$

where τ is a temperature parameter, and the expectation is taken over the stochastic dual-view augmentation process.

By enforcing agreement between prediction distributions under weak and strong perturbations, DVPC provides a label-independent stabilizing signal that mitigates overfitting to noisy labels and enhances prediction reliability. This consistency regularization complements the supervised objective by constraining the prediction space independently of the observed noisy labels.

D. Cross-Modal Representation Contrast

To promote cross-modal alignment and cross-view invariance, we construct contrastive pairs using the projected HSI and LiDAR features from weakly and strongly augmented views. Let $Z_v^m \in \mathbb{R}^{N\times d_p}$ denote the matrix of projected features for modality $m \in \{h, l\}$ and view $v \in \{w, s\}$, where N is the minibatch size and $d_p = 32$ is the projection dimension. We define four ordered feature-set pairs

$$\mathcal{P}_{\text{cm}} = \left\{\left(Z_w^h, Z_w^l\right), \left(Z_s^h, Z_s^l\right), \left(Z_w^h, Z_s^h\right), \left(Z_w^l, Z_s^l\right)\right\}.$$

The first two pairs align HSI and LiDAR representations within each augmented view, while the latter two encourage consistency between weak and strong views within each modality.

For an ordered pair $(A, B) \in \mathcal{P}_{\text{cm}}$, let a_i and b_i denote the corresponding representations of the same pixel. The directional InfoNCE loss is defined as follows:

$$\mathcal{L}_{\text{InfoNCE}}(A, B) = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(\text{sim}\left(a_i, b_i\right)/\tau_m\right)}{\sum_{j=1}^{N} \exp\left(\text{sim}\left(a_i, b_j\right)/\tau_m\right)} \tag{10}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, implemented as the dot product of ℓ_2-normalized features, and $\tau_m = 0.5$ is the temperature parameter. For each anchor a_i, the denominator includes its positive counterpart b_i and the remaining $N - 1$ representations from B, which are treated as negatives.

The CMRC objective averages the losses over the four ordered feature-set pairs

$$\mathcal{L}_{\text{cmrc}} = \frac{1}{4}\sum_{(A,B)\in\mathcal{P}_{\text{cm}}} \mathcal{L}_{\text{InfoNCE}}(A, B). \tag{11}$$

Each term uses representations from the first feature set as anchors and those from the second as candidates, without an explicit reverse-direction term. Together, these objectives encourage cross-modal correspondence and intramodal view consistency through label-independent contrastive regularization.

E. View-Invariant Representation Contrast

Beyond the cross-modal alignment promoted by CMRC, robust pixel-level discrimination under noisy supervision requires the representation of the same pixel to remain stable under different perturbation strengths. To this end, we introduce a view-invariant representation contrastive objective that enforces instance-level consistency between weakly and strongly augmented views at both unimodal and fused representation levels, following the symmetric NT-Xent formulation in [45].

Let z_v^h and z_v^l denote the projected HSI and LiDAR embeddings obtained from view $v \in \{w, s\}$, respectively, and let f_v denote the corresponding fused feature without an additional projection head. For notational convenience, the representation of pixel i in feature space $t \in \{h, l, f\}$ is defined as follows:

$$\mathbf{r}_{v,i}^t = \begin{cases} z_{v,i}^h, & t = h \\ z_{v,i}^l, & t = l \\ f_{v,i}, & t = f. \end{cases} \tag{12}$$

For each feature space, the weak and strong representations of the same pixel constitute a positive pair. All representations of the other pixels from both views are treated as negative samples. Let $\bar{v}$ denote the complementary view of v, such that $\bar{w} = s$ and $\bar{s} = w$. For an anchor $\mathbf{r}_{v,i}^t$, its positive sample is therefore $\mathbf{r}_{\bar{v},i}^t$. To simplify the formulation, we first define the normalization term as follows:

$$D_{v,i}^t = \sum_{\substack{u\in\{w,s\},\, 1\le j\le N \\ (u,j)\neq(v,i)}} \exp\left(\frac{\text{sim}\left(\mathbf{r}_{v,i}^t, \mathbf{r}_{u,j}^t\right)}{\tau_v}\right) \tag{13}$$

where N is the minibatch size and τ_v is the temperature parameter. The summation excludes only the anchor itself.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

Therefore, $D^t_{v,i}$ contains the positive counterpart $\mathbf{r}^t_{\bar{v},i}$ and the remaining $2N-2$ negative representations.

The contrastive loss associated with each anchor is then written as follows:

$$\ell^t_{v,i} = -\log \frac{\exp\left(\operatorname{sim}\left(\mathbf{r}^t_{v,i}, \mathbf{r}^t_{\bar{v},i}\right) \Big/ \tau_v\right)}{D^t_{v,i}}. \tag{14}$$

By treating representations from both views as anchors, the view-invariant loss in feature space t is defined as follows:

$$\mathcal{L}^t_{\text{virc}} = \frac{1}{2N} \sum_{v\in\{w,s\}} \sum_{i=1}^{N} \ell^t_{v,i}, \qquad t \in \{h, l, f\}. \tag{15}$$

The complete VIRC objective aggregates the losses from the HSI, LiDAR, and fused feature spaces

$$\mathcal{L}_{\text{virc}} = \sum_{t\in\{h,l,f\}} \mathcal{L}^t_{\text{virc}} \tag{16}$$

where $\operatorname{sim}(\cdot,\cdot)$ denotes cosine similarity

$$\operatorname{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}. \tag{17}$$

Unlike a directional weak-to-strong contrastive objective, VIRC treats the representations from both augmented views as anchors and thus enforces weak-to-strong and strong-to-weak consistency symmetrically. At the unimodal level, it improves the perturbation invariance of the HSI and LiDAR representations. At the fused level, it stabilizes the joint multimodal representation. By jointly promoting intrainstance compactness and interinstance discrimination, VIRC enhances robustness to augmentation-induced variations and noisy supervision.

F. Total Objective

Let $\tilde{y}$ denote the observed noisy label for each training pixel. The supervised learning objective is formulated using a cross-entropy loss computed on the weakly augmented view. In practice, we further incorporate mixup [48] on weak-view samples to alleviate overfitting to corrupted labels. We denote the resulting supervised loss by $\mathcal{L}_{sup}$.

The final training objective integrates the proposed consistency and contrastive regularization terms

$$\mathcal{L}_{\text{total}} = \mathbb{E}\left[\mathcal{L}_{\text{sup}} + \alpha\, \mathcal{L}_{\text{dvpc}} + \beta\, \mathcal{L}_{\text{cmrc}} + \gamma\, \mathcal{L}_{\text{virc}}\right] \tag{18}$$

where α, β, and γ are weighting coefficients that control the contributions of DVPC, CMRC, and VIRC, respectively. All supervised losses are computed using predictions from the weak view, while consistency and contrastive regularization exploit both weak and strong views. This unified objective enables robust multimodal representation learning by jointly leveraging supervision based on noisy labels and label-independent regularization.

G. Computational Complexity Analysis

The proposed DCCL framework introduces additional consistency objectives with moderate computational overhead. Let d_f, d_p, K, and B denote the fused-feature dimension, projection dimension, number of classes, and minibatch size, respectively. Let F_{enc} denote the feature-extraction cost for one sample. The weak and strong views are processed by the same modality-specific encoders and therefore do not require duplicated encoder parameters. Processing ν views incurs a minibatch forward cost of $O(\nu B F_{\text{enc}})$, with $\nu = 2$ for the weak and strong views. The backward pass has the same asymptotic order. When the additional mixup branch used in our training implementation is included, the number of forward passes becomes three.

DVPC computes the KL divergence between two K-dimensional prediction distributions and therefore has a computational complexity of $O(BK)$. CMRC constructs cross-modal and cross-view similarity matrices from projected embeddings, resulting in $O(B^2 d_p)$ complexity. VIRC performs instance-level contrastive comparisons over the fused and modality-specific representations, resulting more precisely in $O(B^2(d_f + 2d_p))$, which can be simplified to $O(B^2 d_p)$ when the representation dimensions are of the same order. These objectives require $O(B^2)$ auxiliary memory for the pairwise similarity matrices. Since the representation dimensions are kept relatively low, the additional overhead remains manageable. Accordingly, excluding the optional mixup branch and omitting constant factors, the per-iteration training complexity is

$$O(\nu B F_{\text{enc}}) + O\left(B^2\left(d_f + d_p\right)\right) + O(BK). \tag{19}$$

IV. Experiments

A. Datasets

To comprehensively evaluate the effectiveness of the proposed method on multimodal HSI classification, three widely used benchmark datasets are adopted in our experiments, namely Houston 2013, Trento, and MUUFL Gulfport. All these datasets provide coregistered HSI and LiDAR data, which are commonly used for validating HSI–LiDAR fusion methods. Table I summarizes the class information and the numbers of training and testing samples for the Houston 2013, Trento, and MUUFL datasets. For all datasets, 50 labeled samples per class are used for training, while the remaining samples are reserved for testing. The three datasets are described in detail below.

1) *Houston 2013 Dataset:* The Houston 2013 dataset, released for the IEEE GRSS Data Fusion Contest, consists of coregistered HSI and LiDAR data acquired over an urban area. The HSI was captured by the CASI sensor with 144 spectral bands and a spatial resolution of 2.5 m, while the LiDAR data provide elevation information at the same resolution. The dataset contains 15 land-cover classes and is widely used for evaluating multimodal urban classification methods.
2) *Trento Dataset:* The Trento dataset was collected over a rural area in Trento, Italy, and it includes HSI and corresponding LiDAR data. The HSI data were acquired by the AISA Eagle sensor and contain 63 spectral bands covering approximately 0.42–0.99 μm, with a spatial resolution of 1 m. It contains six land-cover classes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

dominated by vegetation types, making it suitable for assessing the complementarity of spectral and elevation information.

3) *MUUFL Dataset*: The MUUFL Gulfport dataset [49], [50] consists of coregistered HSI and LiDAR data acquired over an urban–coastal environment. The HSI was captured by the CASI-1500 sensor with 64 spectral bands, and the LiDAR data provide aligned elevation and intensity information at a spatial resolution of 1 m. The dataset includes 11 land-cover classes and presents challenges such as shadows and mixed pixels.

B. Experimental Details

All experiments were implemented using the PyTorch deep learning framework. The same experimental settings were applied across all three datasets. During training, a patch size of 7×7 was adopted, with a batch size of 64, an initial learning rate of 0.001, and a total of 110 training epochs. The Adam optimizer was used for all experiments. Label noise was introduced only into the training set, while the test labels remained unchanged. All experiments were conducted on a workstation equipped with an Intel Core i7-13700KF CPU (13th generation, 3.4 GHz, 16 cores), 64 GB of system memory, and an NVIDIA GeForce RTX 3090 GPU with 24 GB memory. To reduce randomness, each experiment was repeated five times, and the final performance was reported as the average over five runs. The model obtained at the final training epoch was used for evaluation.

C. Comparison Methods

Existing studies specifically addressing noisy-label learning for HSI–LiDAR classification remain limited. Therefore, we compare the proposed method with five representative HSI–LiDAR multimodal classification methods and two adapted noisy-label learning methods. S2ENet [47] is a spatial–spectral cross-modal enhancement network that explicitly models bidirectional interactions between HSI and LiDAR features before multimodal fusion. HyperMLP [51] is an MLP-Mixer-based multimodal classification framework that integrates HSI and LiDAR data using superpixel priors and feature aggregation strategies. M2FNet [52] combines multiscale 3-D–2-D CNNs with a lightweight attention-free transformer to jointly extract and fuse spectral–spatial and elevation features. CHNet [53] adopts a coarse-to-fine learning strategy that performs feature redundancy reduction and selective fusion to improve the discriminability of multimodal representations. AGMCNet [54] is an attention-guided multiscale convolutional network that decomposes multimodal features and dynamically integrates them through cross-attention-based fusion.

In addition, we adapt two recent noisy-label learning methods to the HSI–LiDAR classification task using S2ENet as their common multimodal backbone. S-directional label diffusion (DLD) combines S2ENet with the DLD model [55], which formulates noisy-label learning as a robust label-generation process and employs directional and random diffusion paths to progressively recover reliable labels from noisy supervision. S-noisy label calibration (NLC) integrates S2ENet with NLC [56], which exploits the complementary information and prediction relationships among multiple views to estimate label confidence and calibrate potentially incorrect labels. For a fair comparison, S2ENet is also employed as the backbone architecture of the proposed method.

For all methods, we used either the official implementations or reproduced them according to the original publications. All methods were evaluated using identical training and test partitions and the same noisy-label realizations. Their main hyperparameters were set according to the original papers and adjusted on the training data when necessary.

D. Results on the Houston 2013 Dataset

Table II reports the quantitative results on the Houston 2013 dataset under 40% symmetric and asymmetric label noise, while Figs. 2 and 3 show the corresponding classification maps.

Under symmetric label noise, the proposed method achieves the best overall performance, with an OA of 93.37%, an AA of 94.46%, and a Kappa coefficient ($\times 100$) of 92.83. Compared with the best competing result for each metric, the improvements are 7.11, 6.01, and 7.69 percentage points, respectively. Our method achieves the highest accuracy for stressed grass, residential, railway, and parking lot 1 and obtains a tied best accuracy of 100% for synthetic grass. Particularly, it outperforms the strongest competitors by 7.10 and 9.16 percentage points for residential and parking lot 1, respectively. Although some methods perform well on individual classes, their results vary considerably across categories. On the contrary, our method provides more balanced class-wise performance and low standard deviations, demonstrating its robustness to randomly corrupted labels.

Under asymmetric label noise, the proposed method obtains an OA of 82.15%, an AA of 84.39%, and a Kappa coefficient ($\times 100$) of 80.71, exceeding the best competing results by 9.91, 12.66, and 10.73 percentage points, respectively. It achieves the highest accuracies for seven classes, including stressed grass, soil, water, road, railway, parking lot 1, and tennis court. Notable improvements are observed for water and parking lot 1, with gains of 10.96 and 16.17 percentage points, respectively. These results indicate that the proposed method can effectively alleviate class-dependent supervision bias and maintain balanced performance under structured label corruption.

The qualitative results in Figs. 2 and 3 are consistent with the quantitative comparisons. As highlighted by the red bounding boxes, our method better preserves the shapes and spatial distributions of small and regularly structured objects, particularly the athletic-field facilities and surrounding regions. Compared with the competing methods, it produces more complete object regions and fewer fragmented or isolated misclassifications, especially under asymmetric noise.

Overall, the results on Houston 2013 demonstrate that the proposed method is robust to both random and class-dependent label corruption, particularly in preserving small and regularly structured urban objects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

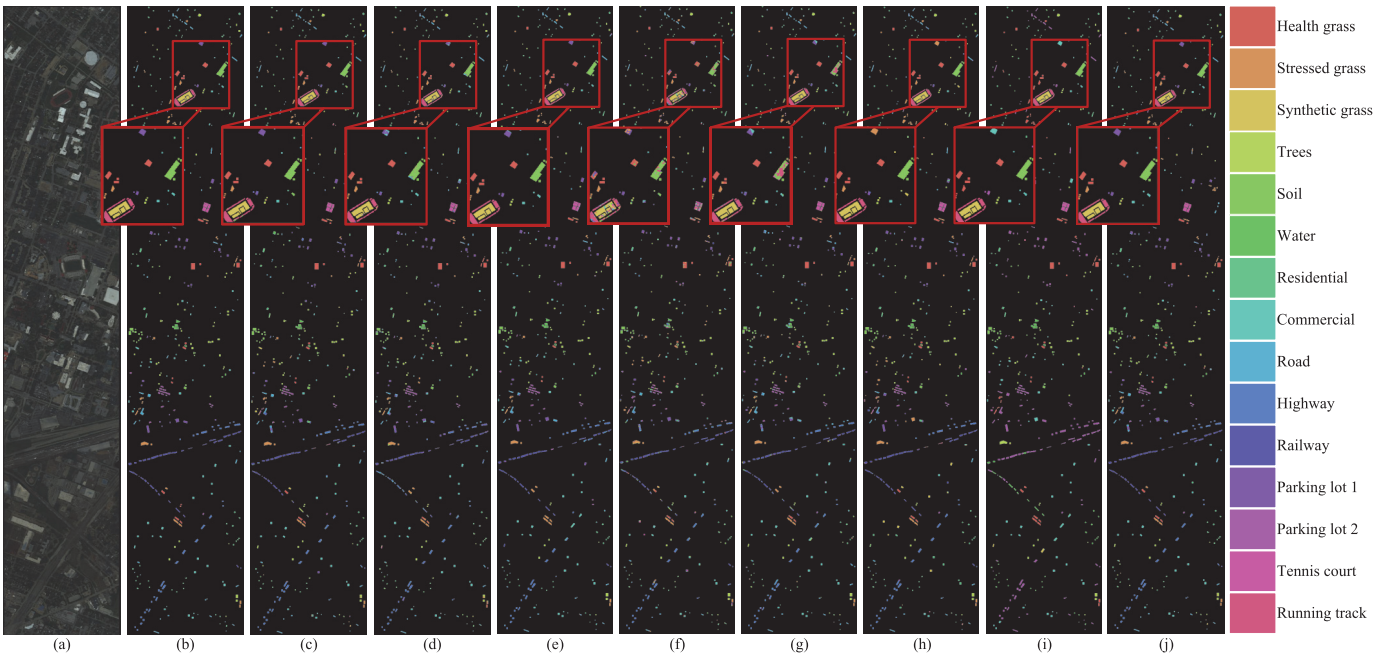


Fig. 2. Performance comparison on the Houston 2013 dataset under 40% symmetric label noise. (a) False color image. (b) Ground truth. (c) S2ENet. (d) HyperMLP. (e) M2FNet. (f) CHNet. (g) AGMCNet. (h) S-DLD. (i) S-NLC. (j) Ours.

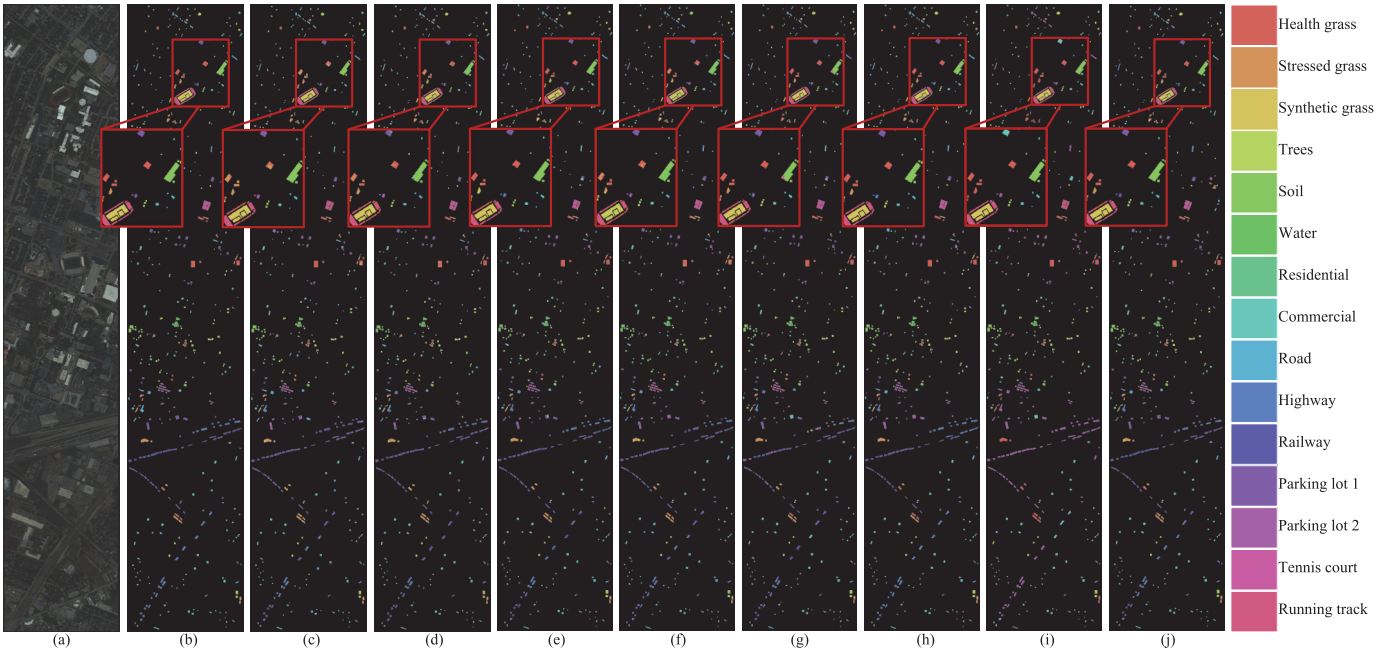


Fig. 3. Performance comparison on the Houston 2013 dataset under 40% asymmetric label noise. (a) False color image. (b) Ground truth. (c) S2ENet. (d) HyperMLP. (e) M2FNet. (f) CHNet. (g) AGMCNet. (h) S-DLD. (i) S-NLC. (j) Ours.

E. Results on the Trento Dataset

Table III reports the quantitative results on the Trento dataset under 40% symmetric and asymmetric label noise, while Figs. 4 and 5 show the corresponding classification maps.

Under symmetric label noise, the proposed method achieves the best overall performance, with an OA of 95.67%, an AA of 94.02%, and a Kappa coefficient ($\times 100$) of 94.24. Compared with the best competing result for each metric, the improvements are 2.98, 6.21, and 4.09 percentage points, respectively. At the class level, our method achieves the highest accuracies for apple trees and wood, reaching 96.63% and 98.98%, respectively. Although some competing methods obtain higher accuracies for individual classes, their performance varies considerably across categories. On the contrary, our method maintains consistently high accuracies across all six classes, demonstrating its robustness to randomly corrupted labels.

Under asymmetric label noise, the proposed method obtains an OA of 86.76%, an AA of 85.46%, and a Kappa coefficient

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

TABLE II

PER-CLASS ACCURACY (%) ON THE HOUSTON 2013 DATASET UNDER 40% SYMMETRIC AND ASYMMETRIC LABEL NOISE. MEAN VALUES ARE REPORTED IN THE FIRST ROW AND STANDARD DEVIATIONS IN THE SECOND ROW

Class	Symmetric noise								Asymmetric noise							
	S2ENet	HyperMLP	M2FNet	CHNet	AGMCNet	S-DLD	S-NLC	**Ours**	S2ENet	HyperMLP	M2FNet	CHNet	AGMCNet	S-DLD	S-NLC	**Ours**
1	91.43	86.34	86.63	77.52	82.66	96.34	**98.09**	97.12	66.85	70.92	77.59	59.93	66.63	31.89	**95.48**	85.06
	(9.88)	(4.11)	(6.68)	(1.62)	(11.96)	(0.56)	(0.00)	(0.95)	(27.96)	(7.92)	(7.48)	(3.10)	(20.04)	(39.20)	(4.74)	(0.70)
2	89.58	82.86	93.85	78.17	82.66	90.20	0.00	**95.07**	84.64	83.52	84.29	71.40	80.47	87.33	33.10	**94.15**
	(5.49)	(5.89)	(3.96)	(2.67)	(11.96)	(3.24)	(0.00)	(1.44)	(12.88)	(5.41)	(10.94)	(2.21)	(13.70)	(8.13)	(36.45)	(1.98)
3	99.90	97.93	99.72	78.15	93.08	**100.00**	**100.00**	**100.00**	96.75	98.49	92.46	65.50	93.08	99.36	99.63	**97.31**
	(0.13)	(1.02)	(0.18)	(1.06)	(8.97)	(0.00)	(0.00)	(0.00)	(2.09)	(0.86)	(8.58)	(1.91)	(8.97)	(0.42)	(0.58)	(2.67)
4	76.49	97.72	90.87	86.36	70.59	97.76	**98.58**	97.29	65.33	81.79	76.63	63.22	70.59	93.09	**99.13**	86.43
	(19.77)	(1.83)	(4.39)	(1.56)	(19.09)	(0.30)	(0.42)	(1.20)	(16.28)	(8.77)	(13.70)	(3.32)	(19.09)	(6.10)	(0.42)	(4.16)
5	83.15	92.67	96.81	81.76	86.78	**99.98**	60.00	99.93	80.19	84.24	84.82	67.45	80.94	71.54	79.98	**92.75**
	(19.72)	(2.04)	(2.78)	(1.38)	(9.31)	(0.03)	(48.99)	(0.15)	(9.87)	(5.85)	(6.54)	(3.57)	(13.48)	(17.63)	(39.99)	(5.81)
6	93.96	89.75	90.62	58.76	81.45	95.64	**100.00**	99.78	82.40	69.60	53.16	61.09	80.36	53.96	83.73	**94.69**
	(2.21)	(2.05)	(6.77)	(1.47)	(7.40)	(1.15)	(0.00)	(0.33)	(7.14)	(5.88)	(21.66)	(6.08)	(6.72)	(9.45)	(15.63)	(2.15)
7	76.37	82.09	86.40	63.52	79.33	82.86	81.87	**93.50**	61.52	62.09	79.36	56.78	66.09	21.22	**98.96**	70.84
	(6.23)	(4.11)	(4.76)	(4.57)	(6.96)	(4.72)	(22.28)	(3.02)	(8.79)	(5.80)	(6.49)	(4.85)	(16.34)	(18.64)	(0.98)	(4.59)
8	68.85	76.42	75.28	58.78	71.31	29.03	**83.32**	81.49	53.46	64.76	66.60	55.36	65.70	68.48	**86.05**	74.87
	(7.23)	(2.63)	(4.74)	(1.80)	(10.56)	(1.84)	(10.41)	(4.42)	(11.43)	(11.65)	(12.20)	(3.13)	(16.58)	(22.60)	(8.59)	(8.62)
9	69.41	78.10	72.33	70.28	**82.86**	82.28	0.00	82.10	53.99	59.93	42.60	57.90	43.23	28.51	0.00	**64.73**
	(23.64)	(2.86)	(11.15)	(3.02)	(9.99)	(0.96)	(0.00)	(3.83)	(31.75)	(4.84)	(7.18)	(2.10)	(15.78)	(29.53)	(0.00)	(2.47)
10	77.37	70.13	84.10	80.88	73.19	**96.18**	1.97	93.36	51.55	52.90	**74.10**	59.98	50.91	43.27	0.00	67.71
	(30.19)	(13.87)	(13.07)	(2.13)	(12.37)	(1.20)	(3.94)	(1.68)	(16.73)	(10.30)	(12.26)	(3.10)	(15.19)	(9.61)	(0.00)	(17.08)
11	78.20	71.49	87.34	81.11	72.89	94.90	0.00	**95.54**	60.07	66.97	75.24	58.82	74.03	84.26	0.35	**84.78**
	(32.14)	(2.99)	(6.29)	(2.10)	(8.39)	(1.61)	(0.00)	(1.40)	(20.74)	(14.31)	(10.16)	(3.07)	(12.47)	(9.41)	(0.70)	(6.76)
12	77.21	65.19	78.58	67.61	66.71	69.65	0.00	**87.74**	55.75	42.52	55.94	61.67	51.58	27.24	0.00	**77.84**
	(7.61)	(14.81)	(6.63)	(4.48)	(5.41)	(1.00)	(0.00)	(3.08)	(20.89)	(15.54)	(20.28)	(2.30)	(9.43)	(10.98)	(0.00)	(6.34)
13	89.00	82.82	85.63	68.02	84.73	91.98	**99.02**	95.32	76.91	60.86	61.29	48.69	70.31	74.58	**98.43**	80.95
	(6.25)	(5.04)	(5.55)	(0.00)	(8.81)	(1.34)	(1.95)	(0.82)	(11.89)	(6.39)	(10.27)	(2.70)	(15.83)	(9.22)	(2.02)	(12.17)
14	84.21	90.69	95.87	63.02	83.39	**100.00**	80.00	99.84	58.90	77.83	79.79	63.07	76.93	90.06	40.00	**99.47**
	(20.79)	(4.92)	(1.76)	(2.04)	(10.42)	(0.00)	(40.00)	(0.35)	(23.55)	(11.56)	(11.67)	(2.34)	(16.66)	(9.91)	(48.99)	(0.49)
15	93.64	94.62	84.82	66.62	85.11	**100.00**	**100.00**	98.89	84.14	64.66	72.10	62.49	78.23	98.07	**100.00**	94.33
	(5.92)	(1.23)	(4.57)	(0.67)	(10.09)	(0.00)	(0.00)	(1.53)	(8.70)	(13.05)	(11.19)	(1.95)	(7.19)	(1.24)	(0.00)	(4.93)
OA(%)	81.02	82.28	86.26	73.70	80.12	86.22	50.88	**93.37**	66.28	68.49	72.24	61.17	67.64	60.64	55.51	**82.15**
	(7.72)	(1.01)	(1.51)	(0.82)	(1.74)	(0.30)	(4.59)	(0.56)	(5.19)	(1.49)	(2.26)	(0.46)	(3.89)	(7.38)	(3.79)	(1.56)
AA(%)	83.25	83.92	87.26	72.04	79.78	88.45	60.19	**94.46**	68.83	69.41	71.73	60.89	69.94	64.86	60.99	**84.39**
	(6.38)	(0.76)	(1.02)	(0.70)	(1.76)	(0.27)	(5.16)	(0.47)	(4.95)	(0.63)	(2.62)	(0.29)	(2.75)	(6.00)	(3.45)	(1.39)
κ(×100)	79.52	80.84	85.14	71.99	78.53	85.11	47.87	**92.83**	63.61	65.95	69.98	58.10	65.05	57.51	52.58	**80.71**
	(8.27)	(1.09)	(1.63)	(0.82)	(1.88)	(0.33)	(4.79)	(0.60)	(5.59)	(1.58)	(2.45)	(0.48)	(4.20)	(7.91)	(3.93)	(1.68)

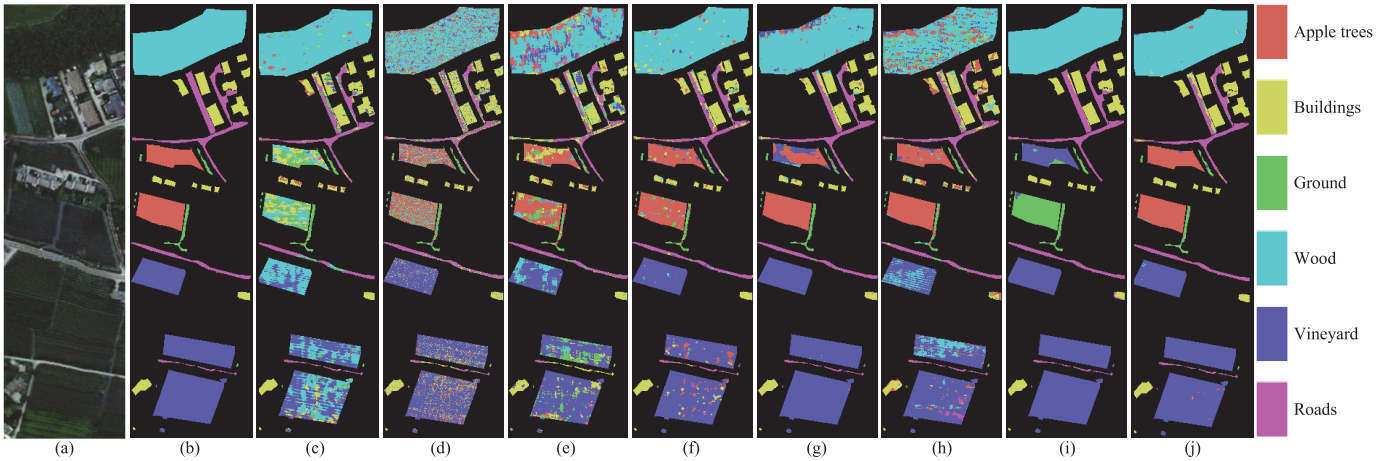


Fig. 4. Performance comparison on the Trento dataset under 40% symmetric label noise. (a) False color image. (b) Ground truth. (c) S2ENet. (d) HyperMLP. (e) M2FNet. (f) CHNet. (g) AGMCNet. (h) S-DLD. (i) S-NLC. (j) Ours.

(×100) of 82.47, exceeding the best competing results by 3.01, 5.35, and 4.03 percentage points, respectively. Some competing methods achieve high accuracies for particular classes but suffer severe degradation in others. For example, S-DLD performs well on apple trees and ground but shows substantially lower accuracies for vineyard and roads, whereas S-NLC achieves high accuracies for buildings, wood, and roads but performs poorly on apple trees and vineyard. By comparison, the proposed method provides more balanced performance across the six categories, indicating its ability to alleviate class-dependent supervision bias.

The qualitative results in Figs. 4 and 5 are consistent with the quantitative findings. Compared with most competing methods, our approach produces more homogeneous

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

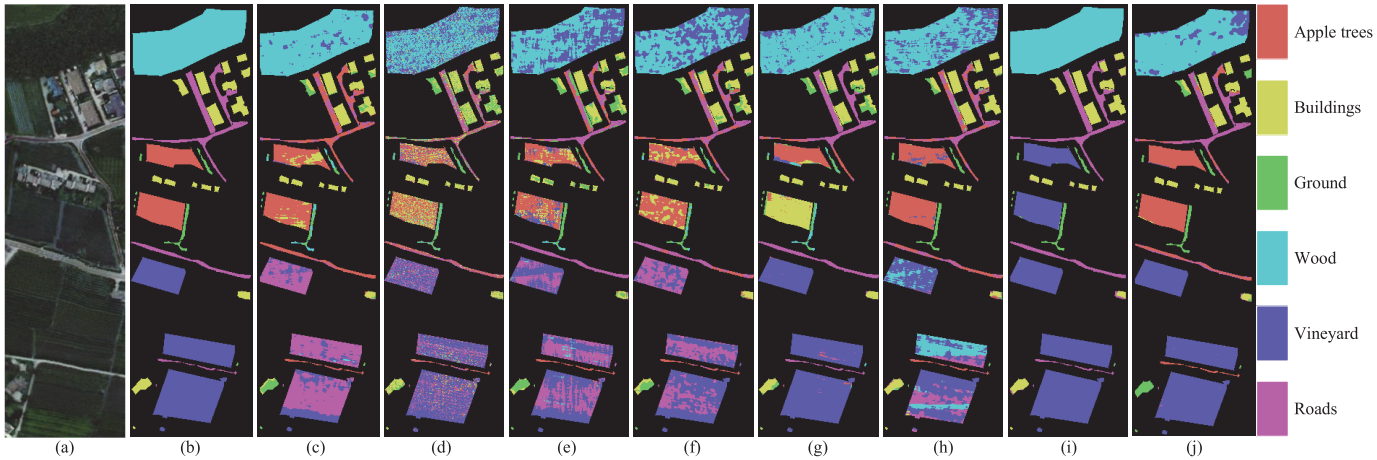


Fig. 5. Performance comparison on the Trento dataset under 40% asymmetric label noise. (a) False color image. (b) Ground truth. (c) S2ENet. (d) HyperMLP. (e) M2FNet. (f) CHNet. (g) AGMCNet. (h) S-DLD. (i) S-NLC. (j) Ours.

TABLE III

PER-CLASS ACCURACY (%) ON THE TRENTO DATASET UNDER 40% SYMMETRIC AND ASYMMETRIC LABEL NOISE. MEAN VALUES ARE REPORTED IN THE FIRST ROW AND STANDARD DEVIATIONS IN THE SECOND ROW

Class	Symmetric noise								Asymmetric noise							
	S2ENet	HyperMLP	M2FNet	CHNet	AGMCNet	S-DLD	S-NLC	**Ours**	S2ENet	HyperMLP	M2FNet	CHNet	AGMCNet	S-DLD	S-NLC	**Ours**
1	44.83	58.75	61.08	88.79	71.63	95.51	0.00	**96.63**	54.25	51.02	56.23	77.88	55.10	**99.00**	19.91	96.32
	(39.82)	(5.01)	(28.70)	(1.74)	(5.84)	(0.28)	(0.00)	(3.42)	(32.08)	(1.30)	(13.88)	(2.27)	(30.26)	(0.19)	(39.83)	(2.07)
2	86.68	81.65	73.75	91.83	96.19	92.95	**97.14**	95.74	67.84	64.08	52.38	69.60	64.96	87.57	**92.60**	72.86
	(10.59)	(2.22)	(5.62)	(0.57)	(1.55)	(1.59)	(0.75)	(1.44)	(10.08)	(3.01)	(5.68)	(1.98)	(16.93)	(1.58)	(2.88)	(3.81)
3	75.06	73.19	70.16	79.77	80.89	80.14	**93.66**	93.52	78.41	62.98	65.27	72.77	70.40	**100.00**	79.30	97.44
	(25.78)	(4.69)	(12.79)	(1.72)	(23.66)	(2.06)	(9.81)	(4.40)	(14.46)	(4.49)	(9.18)	(1.40)	(13.68)	(0.00)	(39.66)	(4.11)
4	74.03	69.94	75.45	95.89	97.19	68.99	59.99	**98.98**	62.71	58.39	52.74	61.43	91.96	75.16	**98.07**	87.74
	(18.43)	(4.57)	(9.78)	(0.40)	(2.97)	(1.68)	(48.98)	(0.73)	(31.27)	(4.91)	(7.04)	(5.52)	(10.09)	(2.89)	(3.83)	(5.38)
5	59.02	69.63	64.68	93.11	**99.80**	93.80	79.93	96.30	36.65	61.21	63.77	64.95	**94.73**	56.86	39.96	91.60
	(29.72)	(6.53)	(20.93)	(1.64)	(0.17)	(1.38)	(39.96)	(4.61)	(17.77)	(4.20)	(7.25)	(5.00)	(4.65)	(5.07)	(48.94)	(7.52)
6	70.79	61.28	56.20	77.16	81.15	80.02	**90.84**	82.93	59.08	48.97	64.28	68.45	78.71	62.06	**95.77**	66.79
	(10.39)	(1.34)	(2.75)	(2.96)	(3.89)	(0.93)	(3.53)	(3.51)	(18.77)	(4.31)	(4.39)	(3.66)	(9.68)	(1.73)	(3.98)	(9.86)
OA(%)	65.78	68.60	67.52	91.40	92.69	84.79	66.21	**95.67**	52.82	58.02	58.41	66.53	83.75	72.12	66.33	**86.76**
	(13.95)	(3.42)	(13.82)	(0.90)	(0.49)	(0.96)	(25.84)	(1.71)	(11.85)	(1.79)	(1.84)	(2.35)	(4.22)	(1.27)	(15.55)	(2.52)
AA(%)	68.40	69.07	66.89	87.76	87.81	85.23	70.26	**94.02**	59.83	57.78	59.11	69.18	75.97	80.11	70.94	**85.46**
	(8.54)	(2.58)	(12.46)	(0.90)	(3.66)	(0.61)	(13.67)	(1.13)	(8.47)	(1.72)	(3.33)	(1.54)	(5.97)	(0.76)	(13.29)	(2.10)
κ(×100)	56.85	59.73	58.82	88.56	90.15	80.24	59.66	**94.24**	40.41	45.83	46.42	56.75	78.44	64.31	56.89	**82.47**
	(15.63)	(4.04)	(16.50)	(1.18)	(0.67)	(1.19)	(26.62)	(2.24)	(14.32)	(2.31)	(2.47)	(2.89)	(5.62)	(1.42)	(18.89)	(3.23)

agricultural parcels and fewer scattered misclassifications. The shapes of the apple orchards and vineyards are more completely preserved, while the narrow road structures are generally more continuous. Under asymmetric noise, some baseline methods produce visually coherent regions but assign entire parcels to incorrect classes. On the contrary, our method reduces such large-area class confusion and generates classification maps that are closer to the ground truth.

Overall, the results on Trento demonstrate that the proposed method maintains balanced class-wise performance and preserves homogeneous agricultural parcels under both symmetric and asymmetric label noise.

F. Results on the MUUFL Dataset

Table IV reports the quantitative results on the MUUFL dataset under 40% symmetric and asymmetric label noise, while Figs. 6 and 7 show the corresponding classification maps.

Under symmetric label noise, the proposed method achieves the best overall performance, with an OA of 80.81%, an AA of 78.45%, and a Kappa coefficient (×100) of 75.39. Compared with the best competing result for each metric, the improvements are 5.38, 6.52, and 6.94 percentage points, respectively. Our method achieves the highest accuracies for six of the 11 classes, including trees, mostly grass, road, building shadow, sidewalk, and yellow curb. Particularly, it outperforms the strongest competitors by 11.00 and 11.39 percentage points for mostly grass and building shadow, respectively. These results demonstrate that the proposed method can effectively mitigate random label corruption and maintain balanced performance across diverse urban classes.

Under asymmetric label noise, the proposed method obtains an OA of 71.51%, an AA of 64.23%, and a Kappa coefficient (×100) of 63.94, exceeding the best competing results by 7.11, 5.53, and 8.48 percentage points, respectively. It achieves the highest accuracies for five classes, namely trees, mostly grass, mixed ground surface, sidewalk, and yellow curb. Although some competing methods perform well on individual categories, their performance varies substantially across classes. On the contrary, the proposed method achieves more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

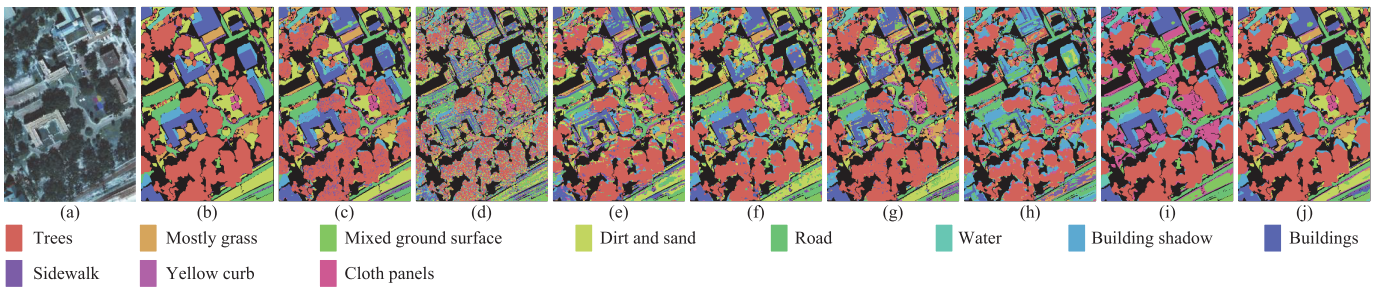


Fig. 6. Performance comparison on the MUUFL dataset under 40% symmetric label noise. (a) False color image. (b) Ground truth. (c) S2ENet. (d) HyperMLP. (e) M2FNet. (f) CHNet. (g) AGMCNet. (h) S-DLD. (i) S-NLC. (j) Ours.

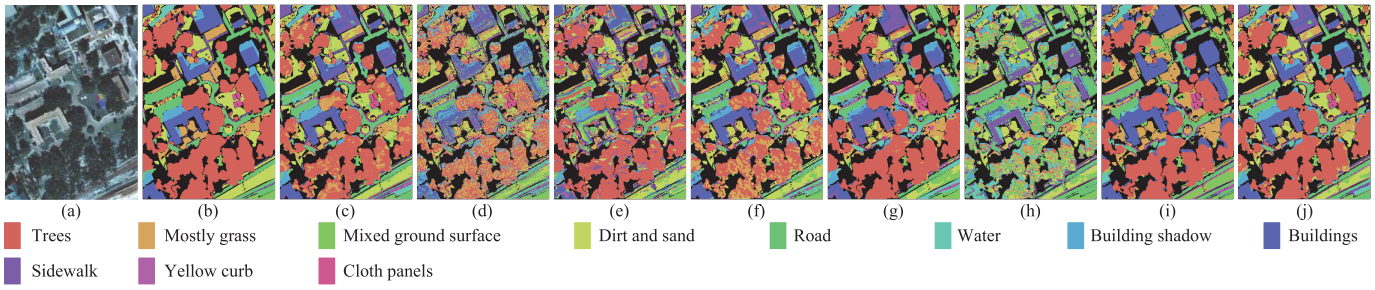


Fig. 7. Performance comparison on the MUUFL dataset under 40% asymmetric label noise. (a) False color image. (b) Ground truth. (c) S2ENet. (d) HyperMLP. (e) M2FNet. (f) CHNet. (g) AGMCNet. (h) S-DLD. (i) S-NLC. (j) Ours.

TABLE IV

PER-CLASS ACCURACY (%) ON THE MUUFL DATASET UNDER 40% SYMMETRIC AND ASYMMETRIC LABEL NOISE. MEAN VALUES ARE REPORTED IN THE FIRST ROW AND STANDARD DEVIATIONS IN THE SECOND ROW

Class	Symmetric noise								Asymmetric noise							
	S2ENet	HyperMLP	M2FNet	CHNet	AGMCNet	S-DLD	S-NLC	**Ours**	S2ENet	HyperMLP	M2FNet	CHNet	AGMCNet	S-DLD	S-NLC	**Ours**
1	65.32	37.66	57.85	72.77	84.26	65.56	77.82	**85.86**	62.43	36.80	59.95	61.00	76.12	63.65	75.70	**82.63**
	(18.94)	(5.52)	(6.83)	(2.89)	(5.79)	(14.95)	(2.13)	(1.97)	(12.59)	(4.24)	(6.26)	(5.42)	(11.95)	(10.62)	(2.42)	(2.86)
2	61.81	35.58	51.71	63.32	60.53	36.52	33.30	**74.32**	42.39	34.42	36.68	46.56	47.55	1.82	50.15	**56.44**
	(8.51)	(3.52)	(10.52)	(4.67)	(11.35)	(24.08)	(40.80)	(3.85)	(22.68)	(2.73)	(3.73)	(1.93)	(16.58)	(2.13)	(41.31)	(7.87)
3	50.67	24.30	49.82	**72.91**	47.85	47.43	0.00	**68.89**	47.15	35.79	52.22	65.48	50.87	57.80	22.53	67.34
	(24.55)	(2.53)	(5.05)	(3.65)	(20.30)	(27.45)	(0.00)	(2.92)	(22.83)	(3.84)	(5.92)	(2.16)	(7.57)	(19.52)	(27.60)	(4.05)
4	66.33	53.42	54.46	80.15	**82.16**	58.77	67.42	70.84	42.21	50.48	47.78	48.89	53.73	48.28	**64.68**	40.91
	(13.30)	(2.44)	(2.23)	(1.20)	(8.52)	(19.00)	(33.71)	(6.37)	(12.57)	(4.47)	(6.18)	(2.34)	(11.30)	(23.57)	(32.60)	(9.54)
5	68.27	39.85	47.32	69.89	77.05	71.73	27.47	**80.71**	43.69	43.08	44.26	58.50	68.24	**71.36**	53.53	65.87
	(13.24)	(2.48)	(4.39)	(2.82)	(6.46)	(11.06)	(34.00)	(3.26)	(22.66)	(3.75)	(8.04)	(0.86)	(12.87)	(6.22)	(30.38)	(6.76)
6	62.93	52.02	58.03	70.82	**92.26**	79.26	72.50	84.09	72.02	49.04	51.39	60.87	**75.05**	63.71	46.88	70.19
	(23.86)	(2.09)	(7.40)	(2.14)	(5.09)	(12.95)	(36.25)	(3.52)	(6.55)	(3.80)	(7.02)	(2.50)	(19.73)	(43.06)	(38.90)	(12.99)
7	62.50	41.04	51.03	68.95	76.74	44.75	62.93	**88.13**	67.32	47.94	53.59	59.23	58.09	68.18	**75.55**	64.37
	(20.42)	(0.78)	(5.05)	(2.96)	(8.16)	(14.00)	(32.00)	(3.40)	(3.82)	(4.09)	(3.20)	(2.15)	(9.03)	(35.04)	(37.80)	(5.68)
8	69.93	28.07	59.31	81.26	**84.47**	69.66	82.24	83.25	56.07	41.87	52.48	67.21	54.86	57.96	**70.64**	65.80
	(10.17)	(2.81)	(5.10)	(2.42)	(7.71)	(17.61)	(15.09)	(1.31)	(9.19)	(2.79)	(3.60)	(1.94)	(5.69)	(27.40)	(14.11)	(4.62)
9	47.43	26.88	38.97	44.63	44.21	15.01	0.00	**62.77**	34.94	30.19	36.10	40.72	28.64	22.88	34.91	**54.97**
	(21.70)	(3.80)	(5.66)	(3.30)	(5.88)	(16.08)	(0.00)	(12.09)	(16.42)	(2.34)	(1.63)	(1.06)	(4.33)	(17.32)	(30.80)	(10.02)
10	63.31	35.04	57.14	68.12	57.89	38.40	43.61	**70.98**	52.78	39.10	50.38	63.61	53.23	50.40	46.62	**65.86**
	(14.07)	(4.96)	(4.96)	(3.28)	(8.76)	(29.71)	(35.89)	(3.42)	(19.24)	(5.48)	(5.23)	(4.05)	(9.76)	(11.48)	(38.54)	(10.72)
11	74.61	65.21	55.25	60.91	83.84	93.88	**97.99**	93.06	70.41	66.48	63.01	73.70	77.99	80.18	**96.16**	72.15
	(15.77)	(6.49)	(3.63)	(1.31)	(14.13)	(4.12)	(2.46)	(1.69)	(9.22)	(4.84)	(5.70)	(1.31)	(2.94)	(13.00)	(1.95)	(2.66)
OA(%)	63.55	35.55	54.31	71.97	75.43	59.97	55.55	**80.81**	54.71	38.83	52.93	59.91	64.40	56.99	61.86	**71.51**
	(14.92)	(2.98)	(3.18)	(1.50)	(1.35)	(11.94)	(7.42)	(0.64)	(7.36)	(1.89)	(2.43)	(2.02)	(3.71)	(8.99)	(7.76)	(0.69)
AA(%)	63.01	39.92	52.81	68.52	71.93	56.45	51.39	**78.45**	53.76	43.20	49.81	58.70	58.58	53.29	57.94	**64.23**
	(12.02)	(1.66)	(1.92)	(0.75)	(1.92)	(6.03)	(10.77)	(0.57)	(4.86)	(1.64)	(0.53)	(0.64)	(3.57)	(6.49)	(10.87)	(3.26)
Kappa(×100)	55.94	25.22	44.47	64.89	68.45	50.35	45.22	**75.39**	45.69	29.13	43.12	51.29	55.46	46.46	53.07	**63.94**
	(15.99)	(2.62)	(3.09)	(1.66)	(1.88)	(13.17)	(8.69)	(0.74)	(7.65)	(1.77)	(2.33)	(1.99)	(3.76)	(11.07)	(8.87)	(0.67)

balanced overall results, indicating its ability to alleviate class-dependent supervision bias in complex urban environments.

The qualitative results in Figs. 6 and 7 are consistent with the quantitative comparisons. Compared with most competing methods, our approach produces more spatially coherent predictions and fewer scattered misclassified pixels in densely structured urban regions. In particular, the distributions of trees and buildings are more completely preserved, while narrow surface classes such as roads and sidewalks exhibit clearer spatial continuity. These advantages remain evident under asymmetric noise, where several baseline methods suffer from severe fragmentation or large-area class confusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

TABLE V

OVERALL ACCURACY (%) UNDER DIFFERENT SYMMETRIC LABEL NOISE RATIOS. ALL RESULTS ARE AVERAGED OVER FIVE INDEPENDENT RUNS, AND THE CORRESPONDING STANDARD DEVIATIONS ARE REPORTED AFTER ±

Dataset	Method	Noise level				
		20%	30%	40%	50%	60%
Houston2013	S2ENet	85.10 ± 7.38	86.82 ± 4.24	81.02 ± 7.72	78.13 ± 4.44	64.84 ± 6.24
	HyperMLP	88.74 ± 0.73	85.13 ± 0.85	82.28 ± 1.01	73.66 ± 0.70	67.76 ± 1.81
	M2FNet	78.09 ± 3.67	73.49 ± 0.82	86.26 ± 1.51	53.42 ± 0.41	43.84 ± 2.20
	CHNet	86.75 ± 0.27	80.23 ± 0.45	73.70 ± 0.82	61.94 ± 1.00	52.21 ± 0.67
	AGMCNet	85.36 ± 0.62	83.10 ± 2.94	80.12 ± 1.74	72.81 ± 3.31	62.03 ± 3.45
	S-DLD	92.79 ± 0.90	92.29 ± 1.99	86.22 ± 0.30	86.39 ± 1.77	82.68 ± 1.03
	S-NLC	51.04 ± 5.06	51.72 ± 3.44	50.88 ± 4.59	40.24 ± 4.86	38.46 ± 3.68
	Ours	**94.74 ± 0.90**	**94.86 ± 0.73**	**93.37 ± 0.56**	**88.00 ± 0.14**	**83.87 ± 1.10**
Trento	S2ENet	73.75 ± 11.95	66.76 ± 22.31	65.78 ± 13.95	70.23 ± 6.70	48.91 ± 16.00
	HyperMLP	85.77 ± 2.06	74.05 ± 2.70	68.60 ± 3.42	49.84 ± 3.99	36.18 ± 2.29
	M2FNet	83.09 ± 4.53	69.58 ± 9.85	67.52 ± 13.82	56.90 ± 2.47	46.58 ± 8.69
	CHNet	95.21 ± 0.84	92.45 ± 1.05	91.40 ± 0.90	70.42 ± 0.60	62.66 ± 2.25
	AGMCNet	97.80 ± 0.41	96.71 ± 0.57	92.69 ± 0.49	84.83 ± 5.51	86.11 ± 5.81
	S-DLD	98.01 ± 0.78	95.13 ± 2.97	84.79 ± 0.96	88.99 ± 5.17	84.72 ± 4.42
	S-NLC	89.63 ± 5.47	91.21 ± 4.37	66.21 ± 25.84	86.93 ± 7.29	54.63 ± 4.97
	Ours	**98.03 ± 0.46**	**96.78 ± 0.22**	**95.67 ± 1.71**	**92.08 ± 1.14**	**90.76 ± 1.07**
MUUFL	S2ENet	72.07 ± 5.55	74.02 ± 2.72	63.55 ± 14.92	58.64 ± 7.92	58.55 ± 5.80
	HyperMLP	59.61 ± 2.41	43.69 ± 2.39	35.55 ± 2.98	25.67 ± 1.66	25.23 ± 1.41
	M2FNet	73.13 ± 1.07	63.82 ± 1.05	54.31 ± 3.18	49.25 ± 2.29	38.27 ± 2.97
	CHNet	81.15 ± 1.26	72.89 ± 2.48	71.97 ± 1.50	65.23 ± 1.00	52.27 ± 9.48
	AGMCNet	80.00 ± 0.67	77.64 ± 2.50	75.43 ± 1.35	69.22 ± 1.63	67.09 ± 2.12
	S-DLD	71.82 ± 2.71	68.67 ± 11.43	59.97 ± 11.94	74.83 ± 4.29	59.47 ± 13.72
	S-NLC	65.54 ± 1.89	59.70 ± 3.76	55.55 ± 7.42	55.78 ± 3.92	53.68 ± 4.32
	Ours	**81.84 ± 0.48**	**81.39 ± 0.63**	**80.81 ± 0.64**	**76.34 ± 1.94**	**76.19 ± 0.20**

Overall, the results on MUUFL demonstrate the robustness of the proposed method in densely structured urban scenes containing irregular objects, mixed pixels, and narrow surface classes.

G. Robustness Under Varying Noise Ratios

To provide a more comprehensive evaluation under different levels of label corruption, we compare all methods under 20%, 30%, 40%, 50%, and 60% symmetric label noise. The corresponding OA results, averaged over five independent runs, are reported in Table V. Performance generally deteriorates as the noise ratio increases, although some methods exhibit nonmonotonic trends.

On Houston 2013, DCCL achieves OA values of 94.74%, 94.86%, 93.37%, 88.00%, and 83.87% under noise ratios from 20% to 60%, respectively. The corresponding improvements over the best competing results are 1.95, 2.57, 7.11, 1.61, and 1.19 percentage points. These results show that DCCL maintains competitive performance across all evaluated noise levels, including the challenging setting in which 60% of the training labels are corrupted.

On Trento, DCCL obtains OA values of 98.03%, 96.78%, 95.67%, 92.08%, and 90.76%. Under 20% and 30% noise, DCCL performs comparably to the strongest competing methods, with small numerical improvements of 0.02 and 0.07 percentage points, respectively. As the noise ratio increases to 40%, 50%, and 60%, its improvements become more evident, reaching 2.98, 3.09, and 4.65 percentage points. This trend indicates that the proposed method becomes particularly beneficial when the available supervision is more severely corrupted.

On MUUFL, DCCL achieves OA values of 81.84%, 81.39%, 80.81%, 76.34%, and 76.19% at the five noise levels. At 20% noise, its performance is comparable to that of CHNet, with a numerical improvement of 0.69 percentage points. At the remaining noise levels, DCCL exceeds the best competing results by 3.75, 5.38, 1.51, and 9.10 percentage points, respectively. In particular, the substantial improvement under 60% noise demonstrates its robustness to severe label corruption in complex urban scenes.

Overall, DCCL achieves the highest numerical OA across all evaluated settings. Under mild noise, it either matches or slightly exceeds the strongest competing methods, while clearer advantages are generally observed under moderate and severe noise. These results demonstrate that the complementary prediction-level and representation-level consistency constraints effectively reduce the model's dependence on corrupted annotations and preserve reliable multimodal information as the noise ratio increases.

H. Ablation Study

To investigate the contribution of each core component, we conduct ablation experiments on the Houston 2013, Trento, and MUUFL datasets under both symmetric and asymmetric label noise. DVPC, CMRC, and VIRC are individually removed from the complete model while all other settings remain unchanged. As reported in Table VI, the complete model consistently achieves the best OA, AA, and Kappa across all datasets and noise conditions. Removing any component results in performance degradation, confirming that all three components contribute positively to noise robustness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

ABLATION STUDY RESULTS ON HOUSTON 2013, TRENTO, AND MUUFL DATASETS UNDER 40% SYMMETRIC AND ASYMMETRIC LABEL NOISE

Dataset	Method	Symmetric noise			Asymmetric noise		
		OA	AA	Kappa	OA	AA	Kappa
Houston2013	Ours	93.37 ± 0.56	94.46 ± 0.47	92.83 ± 0.60	82.15 ± 1.56	84.39 ± 1.39	80.71 ± 1.68
	w/o DVPC	90.57 ± 2.62	92.13 ± 2.11	89.80 ± 2.83	74.80 ± 4.79	76.85 ± 4.53	72.80 ± 5.14
	w/o CMRC	91.55 ± 1.76	92.93 ± 1.44	90.87 ± 1.90	76.02 ± 3.50	78.79 ± 2.76	74.11 ± 3.76
	w/o VIRC	89.88 ± 2.48	91.49 ± 2.24	89.05 ± 2.68	73.59 ± 5.29	75.49 ± 4.85	71.50 ± 5.65
Trento	Ours	95.67 ± 1.71	94.02 ± 1.13	94.24 ± 2.24	86.76 ± 2.52	85.46 ± 2.10	82.47 ± 3.23
	w/o DVPC	91.18 ± 4.85	90.46 ± 3.33	88.46 ± 6.19	80.05 ± 8.88	80.47 ± 5.67	74.46 ± 10.67
	w/o CMRC	93.05 ± 3.43	91.92 ± 1.94	90.78 ± 4.54	81.64 ± 5.37	80.73 ± 5.81	75.76 ± 7.00
	w/o VIRC	76.05 ± 15.37	81.35 ± 8.60	70.33 ± 18.18	76.03 ± 7.51	71.25 ± 6.70	68.10 ± 10.47
MUUFL	Ours	80.81 ± 0.64	78.45 ± 0.57	75.39 ± 0.74	71.51 ± 0.69	64.23 ± 3.26	63.94 ± 0.67
	w/o DVPC	77.09 ± 2.50	75.48 ± 2.18	70.79 ± 3.16	67.68 ± 1.37	59.20 ± 4.05	59.38 ± 1.76
	w/o CMRC	79.43 ± 3.36	77.00 ± 1.90	73.69 ± 4.20	66.16 ± 4.59	59.22 ± 3.73	57.90 ± 4.78
	w/o VIRC	75.57 ± 2.98	75.34 ± 2.04	69.14 ± 3.64	60.16 ± 8.52	59.27 ± 3.90	51.99 ± 8.52

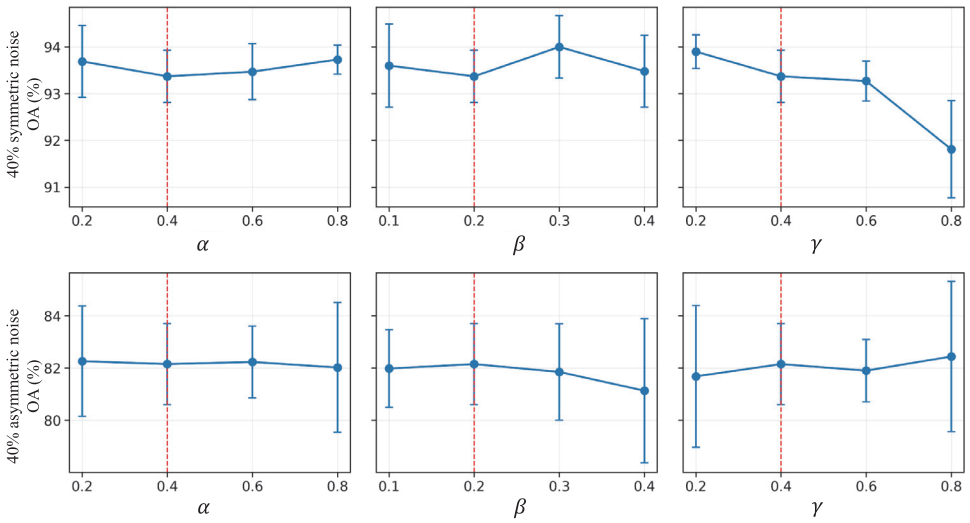


Fig. 8. Sensitivity analysis of the loss-balancing coefficients α, β, and γ on the Houston 2013 dataset under 40% symmetric and asymmetric label noise. The upper and lower rows show the OA results under symmetric and asymmetric noise, respectively. Each point and error bar represents the mean and standard deviation over five runs. The red dashed lines indicate the default setting $(\alpha, \beta, \gamma) = (0.4, 0.2, 0.4)$.

Among the three components, removing VIRC generally causes the most substantial degradation. Across the three datasets, its removal leads to OA reductions of 3.49–19.62 percentage points under symmetric noise and 8.56–11.35 percentage points under asymmetric noise. The performance decreases caused by removing DVPC and CMRC further demonstrate the importance of prediction consistency across augmented views and representation alignment between HSI and LiDAR. Overall, DVPC, CMRC, and VIRC address noisy supervision from complementary prediction-, cross-modal-, and representation-level perspectives, and their joint use effectively improves the robustness of multimodal classification under noisy labels.

I. Sensitivity Analysis of Loss-Balancing Coefficients

We investigate the sensitivity of the loss-balancing coefficients α, β, and γ, which control the contributions of DVPC, CMRC, and VIRC, respectively. The experiments are conducted on the Houston 2013 dataset under 40% symmetric and asymmetric label noise. We evaluate $\alpha, \gamma \in \{0.2, 0.4, 0.6, 0.8\}$ and $\beta \in \{0.1, 0.2, 0.3, 0.4\}$, varying one coefficient at a time while fixing the other two at their default values, i.e., $(\alpha, \beta, \gamma) = (0.4, 0.2, 0.4)$. All configurations use identical data partitions and training settings, and the results are reported as the mean and standard deviation over five runs.

As shown in Fig. 8, DCCL exhibits relatively low sensitivity to α and β within the tested ranges. Under symmetric label noise, the OA ranges from 93.37% to 93.73% as α varies from 0.2 to 0.8, and from 93.37% to 94.00% as β varies from 0.1 to 0.4. Under asymmetric label noise, the corresponding OA ranges are 82.02%–82.26% for α and 81.13%–82.15% for β. These small variations suggest that performance is relatively stable across the evaluated nonzero weights of the prediction-level and cross-modal consistency objectives.

In comparison, γ has a more noticeable influence under symmetric label noise. The OA remains between 93.27% and 93.90% for $\gamma \in \{0.2, 0.4, 0.6\}$ but decreases to 91.81% at $\gamma = 0.8$. This suggests that excessive emphasis on view-invariant representation regularization may hinder classification performance. Under asymmetric label noise, increasing γ from 0.4 to 0.8 raises the mean OA slightly from 82.15% to 82.44%, while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

the standard deviation increases from 1.56 to 2.88 percentage points. Thus, the small gain in mean accuracy is accompanied by greater variability across runs.

Overall, the default setting $(\alpha, \beta, \gamma) = (0.4, 0.2, 0.4)$ provides competitive performance under both noise types, although it does not achieve the highest mean OA in every comparison. The results suggest that DCCL tolerates a range of nonzero loss weights, while an excessively large γ can reduce performance under symmetric noise.

V. Conclusion

This article investigates the challenging problem of HSI and LiDAR data classification under noisy supervision. We formulate this problem, establish a baseline, and propose DCCL, a DCCL framework that integrates cross-modal alignment and contrastive regularization to enable robust multimodal representation learning under noisy labels. Extensive experiments on multiple benchmarks demonstrate that the proposed approach consistently outperforms state-of-the-art methods under various noisy-label settings, confirming its effectiveness and robustness. Beyond performance gains, our study reveals that noisy labels pose a significant challenge to HSI and LiDAR data classification. This highlights the substantial, yet largely untapped, potential for modality-aware and noise-robust multimodal learning over traditional, generic fusion schemes. Despite its effectiveness, DCCL introduces additional training overhead due to dual-view processing. The current evaluation also relies on controlled semantic label corruption with coregistered modality inputs and does not establish robustness to observation degradation or geometric misregistration. Future work will investigate empirically grounded class- and instance-dependent label errors, spatially correlated corruption near land-cover boundaries, and the joint influence of sensor degradation, annotation errors, and cross-modal misregistration.

References

[1] Y. Song et al., "Deep learning for hyperspectral image classification: A comprehensive review and future predictions," *Inf. Fusion*, vol. 123, Nov. 2025, Art. no. 103285.

[2] C. Pu, Y. Liu, S. Lin, X. Shi, Z. Li, and H. Huang, "Multimodal deep learning for semisupervised classification of hyperspectral and LiDAR data," *IEEE Trans. Big Data*, vol. 11, no. 2, pp. 821–834, Apr. 2025.

[3] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[4] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data based on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530915.

[5] M. Burke, A. Driscoll, D. Lobell, and S. Ermon, "Using satellite imagery to understand and promote sustainable development," *SSRN Electron. J.*, vol. 371, no. 6535, 2021, Art. no. eabe8628.

[6] X. Zhang, S. Yang, Z. Feng, L. Song, Y. Wei, and L. Jiao, "Triple contrastive representation learning for hyperspectral image classification with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 500116.

[7] L. Peng et al., "Human-annotated label noise and their impact on ConvNets for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1500–1514, 2025.

[8] H. Senaratne, A. Mobasheri, A. L. Ali, C. Capineri, and M. M. Haklay, "A review of volunteered geographic information quality assessment methods," *Int. J. Geographical Inf. Sci.*, vol. 31, no. 1, pp. 139–167, Jan. 2017.

[9] S. Xia, S. Xu, R. Wang, J. Li, and G. Wang, "Building instance mapping from ALS point clouds aided by polygonal maps," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5700813.

[10] X. Li, M. Ding, Y. Gu, and A. Pižurica, "An end-to-end framework for joint denoising and classification of hyperspectral images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3269–3283, Jul. 2023.

[11] B. Tu, C. Zhou, D. He, S. Huang, and A. Plaza, "Hyperspectral classification with noisy label detection via superpixel-to-pixel weighting distance," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4116–4131, Jun. 2020.

[12] Y. Wang et al., "Bidirectional stacking ensemble curriculum learning for hyperspectral image imbalanced classification with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510915.

[13] Y. Gao, S. Xia, C. Wang, X. Xi, B. Yang, and C. Xie, "Semantic segmentation of airborne LiDAR point clouds with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5706614.

[14] W.-T. Zhang, Y. Bai, S.-D. Zheng, J. Cui, and Z.-Z. Huang, "Tensor transformer for hyperspectral image classification," *Pattern Recognit.*, vol. 163, Jul. 2025, Art. no. 111470.

[15] J. A. Gámez García, G. Lazzeri, and D. Tapete, "Airborne and spaceborne hyperspectral remote sensing in urban areas: Methods, applications, and trends," *Remote Sens.*, vol. 17, no. 17, p. 3126, Sep. 2025.

[16] M. Chen et al., "Fractional Fourier-based frequency-spatial–spectral prototype network for agricultural hyperspectral image open-set classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514014.

[17] S.-F. Huang, S.-C. Lu, A. Mukundan, R. Karmakar, and H.-C. Wang, "Application of hyperspectral imaging in environmental monitoring: Air pollution classification and detection," *Opt. Exp.*, vol. 33, no. 8, pp. 17955–17964, 2025.

[18] Z. Zhang et al., "UAV hyperspectral remote sensing image classification: A systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3099–3124, 2025.

[19] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[20] K. Ni, D. Wang, Z. Zheng, and P. Wang, "MHST: Multiscale head selection transformer for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5470–5483, 2024.

[21] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[22] Y. Zhang et al., "Adaptive multi-stage fusion of hyperspectral and LiDAR data via selective state space models," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103488.

[23] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[24] H. Pan et al., "Multiscale split-recombination cooperative fusion network for hyperspectral and LiDAR land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5520124.

[25] B. Yang, X. Wang, Y. Xing, C. Cheng, W. Jiang, and Q. Feng, "Modality fusion vision transformer for hyperspectral and LiDAR data collaborative classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17052–17065, 2024.

[26] S. Liu, T. Yu, J. Zhou, G. Xing, and H. Chen, "Cross-source transformer-based neighborhood contrastive learning for joint classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 124, Dec. 2025, Art. no. 103225.

[27] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[28] Y. He et al., "Multilevel attention dynamic-scale network for HSI and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5529916.

[29] M. Z. U. Rehman, S. M. S. Islam, D. Blake, A. Ulhaq, and N. Janjua, "Deep learning for land use classification: A systematic review of HS-LiDAR imagery," *Artif. Intell. Rev.*, vol. 58, no. 9, p. 272, Jun. 2025.

[30] N. Lvovich Kazanskiy, L. Leonidovich Doskolovich, N. Vladimirovich Golovastikov, and S. Nikolaevna Khonina, "The power of fusion: LiDAR meets hyperspectral imaging in a new era of exploration," *Opt. Laser Technol.*, vol. 192, Dec. 2025, Art. no. 114080.

[31] Q. Song et al., "MCFNet: Multiscale cross-domain fusion network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4703912.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.

[32] D. Kim, Y. Song, H. Kim, O. Kwon, Y.-K. Yeon, and T. Lim, "Airborne multi-seasonal LiDAR and hyperspectral data integration for individual tree-level classification in urban green spaces at city scale," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104319.

[33] T. Liu, B. Hu, Y. Gu, X. Li, and A. Pižurica, "An enhanced classification method based on adaptive multi-scale fusion for long-tailed multispectral point clouds," *Sci. China Inf. Sci.*, vol. 68, no. 8, Aug. 2025, Art. no. 182302.

[34] Y. Jiang, Y. Gu, and X. Li, "Pillar-voxel fusion network for 3D object detection in airborne hyperspectral point clouds," *Sci. China Inf. Sci.*, vol. 69, no. 1, Jan. 2026, Art. no. 112301.

[35] K. Sarpong, M. Awrangjeb, and M. S. Islam, "Dual spectral–spatial residual adaptive network for hyperspectral image classification in the presence of noisy labels," *Eng. Appl. Artif. Intell.*, vol. 142, Feb. 2025, Art. no. 109900.

[36] J. Ma, Y. Zou, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Spatial pooling transformer network and noise-tolerant learning for noisy hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509719.

[37] Y. Xu, T. Peng, S. Wu, R. Su, and X. Niu, "Enhancing label noise robustness for hyperspectral image classification by neighborhood contrastive learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[38] J. Wang et al., "Selective-relaxed contrastive learning for hyperspectral image classification with noisy labels," *Pattern Recognit.*, vol. 171, Mar. 2026, Art. no. 112298.

[39] M. Zhang, Z. Di, H. Liu, F. Jiang, Y. Zhou, and M. Gong, "Uncertainty-aware local Bayesian framework for hyperspectral image classification with noisy labels," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 3, pp. 3055–3069, Mar. 2026.

[40] X. Yue, A. Liu, N. Chen, S. Xia, J. Yue, and L. Fang, "HyperMLL: Toward robust hyperspectral image classification with multisource label learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526515.

[41] S. Ye, D. Chen, S. Han, and J. Liao, "Learning with noisy labels for robust point cloud segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6443–6452.

[42] H. Huang, S. Yuan, C. Wen, Y. Hao, and Y. Fang, "Noisy few-shot 3D point cloud scene segmentation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2024, pp. 11070–11077.

[43] C. Liu, C. M. Albrecht, Y. Wang, and X. Xiang Zhu, "CromSS: Cross-modal pretraining with noisy labels for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4406517.

[44] C. Lu, C. Li, J. Cheng, Y. Ji, G. Chen, and F. Zhang, "Breaking the noise barrier: LLM-guided semantic filtering and enhancement for multi-modal entity alignment," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2025, pp. 33141–33155.

[45] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[46] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 596–608.

[47] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[48] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018, pp. 1–13. [Online]. Available: https://openreview.net/forum?id=r1Ddp1-Rb

[49] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," *University of Florida*, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013.

[50] X. Du and A. Zare, "Scene label ground truth map for MUUFL gulfport data set," *University of Florida*, Gainesville, FL, USA, Tech. Rep. 20170417, 2017.

[51] J. Li, Y. Liu, R. Song, W. Liu, Y. Li, and Q. Du, "HyperMLP: Superpixel prior and feature aggregated perceptron networks for hyperspectral and LiDAR hybrid classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505614.

[52] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2100116.

[53] K. Ni, Y. Xie, G. Zhao, Z. Zheng, P. Wang, and T. Lu, "Coarse-to-fine high-order network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5509716.

[54] Y. Bai, Q. Yang, L. Zhang, T. Zhao, and T. Qi, "AGMCNet: Attention-guided multiscale convolutional network for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 29199–29219, 2025.

[55] S. Hou et al., "Directional label diffusion model for learning from noisy labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 25738–25748.

[56] S. Xu et al., "Noisy label calibration for multi-view classification," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2025, vol. 39, no. 20, pp. 21797–21805.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:40:43 UTC from IEEE Xplore. Restrictions apply.